

FIT1058 — Chapter 3: Proofs

Exercise Solutions

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Exercise 1

Recall the Collatz function

$$\text{Collatz}(x) = \begin{cases} 3x + 1 & \text{if } x \text{ is odd} \\ \frac{x}{2} & \text{if } x \text{ is even} \end{cases}$$

Prove by cases that, for all positive integers n ,

$$\text{Collatz}^{(2)}(n) \leq \frac{3}{2}n + 1.$$

Proof (by cases). Let $\text{Collatz}(x) = C(x)$.

Case 1: x is odd, $x = 2n + 1$.

$$\begin{aligned} C(x) &= 3(2n + 1) + 1 = 6n + 4 = 2(3n + 2) \quad (\text{even}) \\ C^{(2)}(x) &= 3n + 2 = 3\left(\frac{x-1}{2}\right) + 2 = \frac{3x-3}{2} + 2 = \frac{3}{2}x + \frac{1}{2} \\ \implies C^{(2)}(x) &< \frac{3}{2}x + 1. \end{aligned} \tag{1}$$

Case 2: x is even, $x = 2m$. Then $C(x) = m$.

Case 2a: m is odd.

$$C^{(2)}(x) = 3m + 1 = \frac{3}{2}x + 1. \tag{2}$$

Case 2b: m is even.

$$C^{(2)}(x) = \frac{m}{2} = \frac{x}{4} < \frac{3}{2}x < \frac{3}{2}x + 1. \tag{3}$$

By (1), (2), (3), in all cases,

$$C^{(2)}(x) \leq \frac{3}{2}x + 1.$$

■

Exercise 2

A set of strings L is called *hereditary* if it has the following property: for every nonempty string x in L , there is a character in x which can be deleted from x to give another string in L .

Prove by contradiction that every nonempty hereditary set of strings contains the empty string.

Proof (by contradiction). Assume that L is a nonempty hereditary set of strings that does *not* contain the empty string.

Let $S = \{|x| : x \in L\}$ be the set of lengths of strings in L . Since L is nonempty, S is a nonempty subset of $\mathbb{N}$. By the Well-Ordering Principle, S has a smallest element.

Let $x_0 \in L$ be a string of minimum length, i.e.

$$\forall x \in L, |x_0| \leq |x|.$$

Since x_0 is a nonempty string, its length is $|x_0| = n_0 \geq 1$.

Since L is hereditary and $x_0 \in L$ is nonempty, there is a character in x_0 which can be deleted to give another string $x_1 \in L$, where

$$|x_1| = n_0 - 1.$$

Since x_0 is the smallest string in L , we have $|x_0| \leq |x|$ for every $x \in L$. But

$$|x_1| = n_0 - 1 < n_0 = |x_0|,$$

which contradicts the minimality of x_0 .

This is a contradiction. Therefore, the assumption is false, and L must contain the empty string. ■

Exercise 5

Let E be a finite set and let $\mathcal{I}$ be a collection of subsets of E that satisfies the following three properties:

- (I1) $\emptyset \in \mathcal{I}$.
- (I2) Whenever $X \subseteq Y$ and $Y \in \mathcal{I}$, we also have $X \in \mathcal{I}$. This is called the *hereditary property*.
- (I3) Whenever $X, Y \in \mathcal{I}$ and $|X| < |Y|$, there exists $y \in Y \setminus X$ such that $X \cup \{y\} \in \mathcal{I}$. In other words, given two different-sized sets in $\mathcal{I}$, you can always enlarge the smaller set, using some new member from the larger set, to get another set in $\mathcal{I}$. This is called the *independence augmentation property*.

The sets in $\mathcal{I}$ are called *independent*.

For example, if $E = \{a, b, c\}$, then each of the following collections satisfies all three properties (I1)–(I3):

- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}\}$
- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, c\}, \{b, c\}\}$
- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}\}$
- $\mathcal{I} = \{\emptyset\}$

By way of counterexample, using the same E , each of the following collections fails to satisfy all three properties (I1)–(I3):

- $\mathcal{I} = \{\{a\}, \{b\}, \{a, b\}\}$. This fails (I1) and (I2).
- $\mathcal{I} = \{\emptyset, \{a\}, \{a, b\}\}$. This fails (I2).

- $\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}\}$. This satisfies (I1) and (I2) but fails (I3).
 - $\mathcal{I} = \emptyset$. This fails (I1).
- (a) For each counterexample above that fails (I2) and/or (I3), give specific sets X and Y for which the property fails.
- (b) Prove that, for a social network, the set of cliques satisfies (I1) and (I2) but not, in general, (I3).
- (c) Prove by contradiction that all maximal independent sets have the same size.

Suppose that each member $e \in E$ has a nonnegative real weight $w(e)$, so $w : E \rightarrow \mathbb{R}_0^+$. We use this weight function to also give weights to subsets of E , by defining the weight of a set to be the sum of the weights of its elements: for all $X \subseteq E$, define

$$w(X) := \sum_{e \in X} w(e).$$

Suppose we seek an independent set of maximum weight. Consider the following *greedy algorithm*, so called because it always makes the choice that gives biggest immediate gain:

Algorithm 1: Greedy algorithm for maximum weight independent set

Input: a finite set E , a function $w : E \rightarrow \mathbb{R}_0^+$, a collection $\mathcal{I}$ of subsets of E

Output: an independent set X

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1  $X \leftarrow \emptyset$ 
2  $i \leftarrow 1$ 
3 while  $\exists y \in E \setminus X$  such that  $X \cup \{y\} \in \mathcal{I}$  do
4   choose  $y$  with largest weight  $w(y)$  among all such candidates
5    $e_i \leftarrow y$ 
6    $X \leftarrow X \cup \{e_i\}$ 
7    $i \leftarrow i + 1$ 
8 return  $X$ 
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- (d) Construct a simple social network, and give a nonnegative real weight to each person, such that the greedy algorithm, with $\mathcal{I}$ being the set of all cliques, does not find the maximum weight clique.

Now suppose $\mathcal{I}$ satisfies (I1)–(I3). The greedy algorithm chooses elements of E in a specific order, represented as $e_1, e_2, \dots, e_r$.

- (e) Prove by contradiction that the weights of the chosen elements are decreasing, i.e.,

$$w(e_1) \geq w(e_2) \geq \dots \geq w(e_r).$$

(“Decreasing” allows the possibility that some weights stay the same; we don’t require strictly decreasing.)

- (f) Prove that the greedy algorithm finds a maximum weight independent set.

Solution. (a) ■

Solution. (b)

Proof of (b). Let E = vertices of a social network, and let $\mathcal{I}$ = the set of all cliques.

(I1). Let X be a clique, and consider $X = \emptyset$. Since $\emptyset$ contains no elements, there do not exist $u, v \in \emptyset$ with $u \neq v$.

So the implication “ $u \neq v \implies (u, v)$ is edge”

holds vacuously for all $u, v \in \emptyset$. So $\emptyset$ is a clique, i.e.

$$\emptyset \in \mathcal{I}.$$

(I2). Need to prove: if $X \subseteq Y$ and Y is a clique, then X is a clique.

For all $u, v \in X$ with $u \neq v$: since $X \subseteq Y$,

$$u, v \in Y \quad \text{with } u \neq v.$$

Since Y is a clique, (u, v) is an edge. Hence X is also a clique.

(I3) fails in general. Consider the graph with vertices $\{a, b, c, d\}$ and a single edge (b, c) . Let

$$X = \{a\}, \quad Y = \{b, c\}.$$

X and Y are cliques, and $Y \setminus X = \{b, c\}$. Checking each candidate:

$$X \cup \{b\} = \{a, b\} \text{ is not a clique, since } (a, b) \text{ is not an edge.}$$

$$X \cup \{c\} = \{a, c\} \text{ is not a clique, since } (a, c) \text{ is not an edge.}$$

So no valid $y \in Y \setminus X$ exists, and (I3) fails. ■

Remark. This is the key place where the “clique matroid” fails to be a genuine matroid: in the vector-space analogy, (I3) is the exchange axiom that guarantees any linearly independent set can be extended toward a larger independent set. Cliques do not have this exchange property in general, which is exactly why the greedy algorithm can fail on them (part (d)). ■

Solution. (c)

Proof of (c). An independent set $X \in \mathcal{I}$ is *maximal* if

$$\nexists Y \in \mathcal{I} \text{ such that } X \subsetneq Y.$$

We need to prove:

$$\forall X, Y \in \mathcal{I}, \quad X \text{ maximal} \wedge Y \text{ maximal} \implies |X| = |Y|.$$

Assume, for contradiction, there exist two maximal independent sets $X, Y \in \mathcal{I}$ with $|X| \neq |Y|$. WLOG, assume

$$|X| < |Y|.$$

Since $X, Y \in \mathcal{I}$ and $|X| < |Y|$, by (I3) there exists $y \in Y \setminus X$ such that

$$X \cup \{y\} \in \mathcal{I}.$$

Since $y \notin X$, we have

$$X \subsetneq X \cup \{y\}, \quad X \cup \{y\} \in \mathcal{I}.$$

This contradicts the assumption that X is maximal.

Therefore, the assumption is false: all maximal independent sets have the same size. ■

Remark. This is the exact analogue of the fact that any two bases of a vector space have the same size (i.e. dimension is well defined). There, the argument uses the Steinitz exchange lemma; here (I3) plays precisely that role. ■

Solution. (d) ■

Solution. (e)

Proof of (e). When $\mathcal{I}$ satisfies (I1)–(I3), the greedy algorithm chooses elements of E in a specific order $e_1, e_2, \dots, e_r$. We prove

$$w(e_i) \geq w(e_{i+1}) \quad \text{for all } 1 \leq i < r.$$

Recall e_i is the maximum-weight element among all $y \in E \setminus X_{i-1}$ satisfying $X_{i-1} \cup \{y\} \in \mathcal{I}$:

$$X_{i-1} \cup \{e_i\} \in \mathcal{I},$$

$$\forall y \text{ with } X_{i-1} \cup \{y\} \in \mathcal{I}, \quad w(e_i) \geq w(y).$$

Assume, for contradiction, that

$$w(e_i) < w(e_{i+1}) \quad \text{for some } 1 \leq i < r.$$

Since e_{i+1} was chosen by the algorithm at round $i+1$,

$$X_i \cup \{e_{i+1}\} \in \mathcal{I}.$$

Since $X_{i-1} \subseteq X_i$,

$$X_{i-1} \cup \{e_{i+1}\} \subseteq X_i \cup \{e_{i+1}\}.$$

Since $X_i \cup \{e_{i+1}\} \in \mathcal{I}$, by (I2),

$$X_{i-1} \cup \{e_{i+1}\} \in \mathcal{I}.$$

This means e_{i+1} is one of the candidates y satisfying $X_{i-1} \cup \{y\} \in \mathcal{I}$ at round i . Since e_i was chosen at round i as the maximum-weight element among all such candidates,

$$w(e_i) \geq w(e_{i+1}),$$

which contradicts the assumption $w(e_i) < w(e_{i+1})$.

Therefore the assumption is false, and

$$w(e_i) \geq w(e_{i+1}) \quad \text{for all } 1 \leq i < r. \quad \blacksquare$$

Remark. This mirrors a standard fact from linear algebra: if you build up an independent set greedily by always adding the “largest” remaining valid candidate, the sequence of choices is automatically monotone — the exchange property (I2) is what lets you “pull” a later candidate back to an earlier round without breaking independence. ■

Solution. (f)

Proof of (f). Let Y be a maximum weight independent set. By the stopping condition of the greedy algorithm, when $X = \{e_1, \dots, e_r\}$, there is no $y \in E \setminus X$ such that

$$X \cup \{y\} \in \mathcal{I}.$$

So X is a maximal independent set.

By part (c), we know

$$|X| = |Y| = r.$$

Write $Y = \{f_1, \dots, f_r\}$ with

$$w(f_1) \geq w(f_2) \geq \dots \geq w(f_r).$$

Main claim. For each $i = 1, \dots, r$,

$$w(e_i) \geq w(f_i).$$

By contradiction (minimal counterexample): assume there exists a minimum i such that $w(e_i) < w(f_i)$. By minimality of i ,

$$w(e_k) \geq w(f_k) \quad \text{for all } k < i.$$

Let

$$X_{i-1} = \{e_1, \dots, e_{i-1}\}, \quad F_i = \{f_1, \dots, f_i\}.$$

Since $X_{i-1} \subseteq X \in \mathcal{I}$, by (I2), $X_{i-1} \in \mathcal{I}$. Since $F_i \subseteq Y \in \mathcal{I}$, by (I2), $F_i \in \mathcal{I}$. Also

$$|X_{i-1}| = i - 1 < i = |F_i|.$$

By (I3), there exists $y \in F_i \setminus X_{i-1}$ such that

$$X_{i-1} \cup \{y\} \in \mathcal{I}.$$

Since $y \in F_i = \{f_1, \dots, f_i\}$, and these are sorted in decreasing order of weight, f_i has the smallest weight among them, so

$$w(y) \geq w(f_i).$$

Now, since $y \notin X_{i-1}$ and $X_{i-1} \cup \{y\} \in \mathcal{I}$, y is a valid candidate at round i of the greedy algorithm. But e_i was chosen at round i as the element of maximum weight among all such valid candidates, so

$$w(e_i) \geq w(y) \geq w(f_i).$$

This contradicts the assumption $w(e_i) < w(f_i)$. Hence no such i exists, and

$$w(e_i) \geq w(f_i) \quad \text{for all } i = 1, \dots, r.$$

Summing over i :

$$w(X) = \sum_{i=1}^r w(e_i) \geq \sum_{i=1}^r w(f_i) = w(Y).$$

Since Y was assumed to be a maximum weight independent set, $w(X) \geq w(Y)$ forces

$$w(X) = w(Y).$$

Therefore X is also a maximum weight independent set.

$\therefore$ Greedy algorithm finds a maximum weight independent set. ■

Remark. This whole argument is the discrete, combinatorial analogue of the Matroid Greedy Theorem, itself a generalisation of the fact that Gaussian elimination / the Gram–Schmidt-style greedy basis selection process finds an optimal basis with respect to any linear weighting — the same “minimal counterexample + exchange” technique (Steinitz exchange lemma) underlies both. ■

Exercise 10

Consider the following algorithm:

Algorithm 2: Triple-loop counting

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1 for  $i \leftarrow 1$  to  $n$  do
2   for  $j \leftarrow 1$  to  $i$  do
3     for  $k \leftarrow 1$  to  $j$  do
4       [some action]
```

- For each pair i, j , how many times is the action inside the innermost loop performed? Express this in terms of i or j or both.
- For each i , how many times is the action inside the innermost loop performed? Write this as both a sum of some number of terms, and (using a theorem in this chapter) a simple algebraic expression, both in terms of i .
- Now write a sum of some number of terms, in terms of n , for the total number of times the action is performed.
- Work out this expression for $n = 1, 2, 3, 4, 5$, and for as many further values of n as you like.
- Explore how these numbers behave.
- Conjecture a simple algebraic expression, in terms of n , for the total number of times the action is performed.
- Prove, by induction on n , that your expression is correct.

Solution.

- For fixed i, j : j loop from 1 to i , k loop from 1 to j . Observe that for each pair (i, j) , the innermost action is performed j times.
- For fixed i : j loops from 1 to i , and for each j the inner loop runs j times. So the number of times the action is performed is

$$\sum_{j=1}^i j = \frac{i(i+1)}{2}.$$

- Summing over i from 1 to n :

$$\text{num of action} = \sum_{i=1}^n \sum_{j=1}^i j = \sum_{i=1}^n \frac{i(i+1)}{2}.$$

(d) Evaluating $\sum_{i=1}^n \frac{i(i+1)}{2}$ for small n :

$$n = 1 : 1, \quad n = 2 : 4, \quad n = 3 : 10, \quad n = 4 : 20, \quad n = 5 : 35.$$

(e) The sequence $1, 4, 10, 20, 35, \dots$ is the sequence of *tetrahedral numbers*. Each term is obtained from the previous by adding the next triangular number $(1, 3, 6, 10, 15, \dots)$, and the values grow roughly cubically in n .

(f) Conjecture:

$$\sum_{i=1}^n \frac{i(i+1)}{2} = \frac{n(n+1)(n+2)}{6}.$$

Derivation: using $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ and $\sum_{i=1}^n i = \frac{n(n+1)}{2}$,

$$\begin{aligned} \sum_{i=1}^n \frac{i^2 + i}{2} &= \frac{1}{2} \left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right] \\ &= \frac{1}{12} [n(n+1)(2n+1) + 3n(n+1)] \\ &= \frac{1}{12} n(n+1) [(2n+1) + 3] = \frac{1}{12} n(n+1)(2n+4) \\ &= \frac{1}{6} n(n+1)(n+2). \end{aligned}$$

This matches the numerical values in part (d): $n = 5$ gives $\frac{5 \cdot 6 \cdot 7}{6} = 35$. ✓

(g) **Proof (by induction on n).**

Base case ($n = 1$). LHS = $\frac{1 \cdot 2}{2} = 1$. RHS = $\frac{1 \cdot 2 \cdot 3}{6} = 1$. Equal, so the base case holds.

Inductive hypothesis. Assume for some $n \geq 1$,

$$\sum_{i=1}^n \frac{i(i+1)}{2} = \frac{n(n+1)(n+2)}{6}.$$

Inductive step. We show the formula holds for $n+1$:

$$\begin{aligned} \sum_{i=1}^{n+1} \frac{i(i+1)}{2} &= \sum_{i=1}^n \frac{i(i+1)}{2} + \frac{(n+1)(n+2)}{2} \\ &= \frac{n(n+1)(n+2)}{6} + \frac{(n+1)(n+2)}{2} \quad (\text{by inductive hypothesis}) \\ &= \frac{n(n+1)(n+2) + 3(n+1)(n+2)}{6} \\ &= \frac{(n+1)(n+2)(n+3)}{6}. \end{aligned}$$

This matches the conjectured formula with n replaced by $n+1$.

By the Principle of Mathematical Induction, the formula

$$\sum_{i=1}^n \frac{i(i+1)}{2} = \frac{n(n+1)(n+2)}{6}$$

holds for all $n \geq 1$. ■
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Exercise 11

The n -th harmonic number H_n is defined by

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

In this exercise we prove by induction that $H_n \geq \ln(n+1)$. (It follows that the harmonic numbers increase without bound, even though H_n grows more and more slowly as n increases.)

- (i) **Inductive basis:** prove that $H_1 \geq \ln(1+1)$.
- (ii) Let $n \geq 1$. Assume $H_n \geq \ln(n+1)$ (inductive hypothesis). Consider H_{n+1} ; write it recursively using H_n . Use the inductive hypothesis to obtain $H_{n+1} \geq \dots$, then use the fact that $\ln(1+x) \leq x$, and a property of logarithms, to show $H_{n+1} \geq \ln(n+2)$.
- (iii) What can you now conclude, and why?

Advanced afterthoughts.

- The inequality implies $H_n \geq \ln n$, since $\ln(n+1) \geq \ln n$. It is instructive to try to prove this directly by induction — you will probably run into a snag, illustrating that induction sometimes requires proving something stronger than what you set out to prove.
- Would your proof work for logarithms to other bases, apart from e ? Where does the proof use base e ?
- It is known that $H_n \leq (\ln n) + 1$. Can you prove this?

Solution. The n -th harmonic number H_n is defined by

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

Main result: $H_n \geq \ln(n+1)$

Proof (by induction on n). **Base case ($n = 1$).** $H_1 = 1$, and $\ln(1+1) = \ln 2 < 1$. So $H_1 \geq \ln(1+1)$.

Inductive hypothesis. Assume $H_n \geq \ln(n+1)$ for some $n \geq 1$.

Inductive step. We show $H_{n+1} \geq \ln(n+2)$.

By definition,

$$H_{n+1} = H_n + \frac{1}{n+1} \geq \ln(n+1) + \frac{1}{n+1}.$$

We claim $\ln(n+1) + \frac{1}{n+1} \geq \ln(n+2)$. Observe that

$$\ln(n+2) - \ln(n+1) = \ln\left(\frac{n+2}{n+1}\right) = \ln\left(1 + \frac{1}{n+1}\right) \leq \frac{1}{n+1},$$

using $\ln(1+x) \leq x$ (with $x = \frac{1}{n+1}$). Rearranging,

$$\ln(n+1) + \frac{1}{n+1} \geq \ln(n+2).$$

Combining the two inequalities,

$$H_{n+1} \geq \ln(n+1) + \frac{1}{n+1} \geq \ln(n+2).$$

So the statement holds for $n+1$.

Conclusion. By the Principle of Mathematical Induction, $H_n \geq \ln(n+1)$ for all $n \geq 1$. ■

Advanced afterthoughts

1. The inequality $H_n \geq \ln(n+1)$ implies $H_n \geq \ln n$, since $\ln(n+1) \geq \ln n$. Trying to prove $H_n \geq \ln n$ *directly* by induction runs into a snag: in the inductive step, one would need

$$\ln(n+1) - \ln n = \ln\left(1 + \frac{1}{n}\right) \leq \frac{1}{n+1},$$

but $\ln(1+x) \leq x$ only gives the weaker bound $\ln(1 + \frac{1}{n}) \leq \frac{1}{n}$, which is not tight enough (since $\frac{1}{n} > \frac{1}{n+1}$). This illustrates that induction sometimes requires proving something *stronger* than what one set out to prove — the extra “room” in $\ln(n+1)$ compared to $\ln n$ is exactly what makes the induction go through.

2. The proof relies on the inequality $\ln(1+x) \leq x$, which is a property specific to the natural logarithm (it comes from $\ln(1+x)$ being concave with tangent line $y = x$ at $x = 0$). For a logarithm to another base $b \neq e$, the analogous inequality $\log_b(1+x) \leq x$ does not hold in general, so the proof does not transfer directly to other bases without first converting back to $\ln$ via the change-of-base formula.
3. **Claim:** $H_n \leq \ln n + 1$.

Proof (by induction on n). **Base case** ($n = 1$). $H_1 = 1$. $\text{RHS} = \ln 1 + 1 = 0 + 1 = 1$. So $H_1 \leq \ln 1 + 1$.

Inductive hypothesis. Assume $H_n \leq \ln n + 1$ for some $n \geq 1$.

Inductive step. We show $H_{n+1} \leq \ln(n+1) + 1$.

By definition, $H_{n+1} = H_n + \frac{1}{n+1}$. By the inductive hypothesis,

$$H_{n+1} \leq \ln n + 1 + \frac{1}{n+1}.$$

It remains to show $\ln n + \frac{1}{n+1} \leq \ln(n+1)$, i.e.

$$\ln(n+1) - \ln n \geq \frac{1}{n+1}.$$

Using the inequality $\ln x \geq 1 - \frac{1}{x}$ for $x > 0$, take $x = \frac{n+1}{n}$:

$$\ln \left(\frac{n+1}{n} \right) \geq 1 - \frac{n}{n+1} = \frac{1}{n+1}.$$

Since $\ln \left(\frac{n+1}{n} \right) = \ln(n+1) - \ln n$, this gives

$$\ln(n+1) - \ln n \geq \frac{1}{n+1}.$$

Combining:

$$H_{n+1} \leq \ln n + 1 + \frac{1}{n+1} \leq \ln(n+1) + 1.$$

So the statement holds for $n+1$.

Conclusion. By the Principle of Mathematical Induction, $H_n \leq \ln n + 1$ for all $n \geq 1$. ■

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Exercise 12

(Challenge) Let E be a finite set, and let $\mathcal{S}$ be a set of subsets of E . We say $\mathcal{S}$ is *linear under* $\triangle$ if, for every two sets $X, Y \in \mathcal{S}$, we have $X \triangle Y \in \mathcal{S}$.

The *span* of $\mathcal{S}$, written $\text{span}(\mathcal{S})$, is the set of all symmetric differences of any number of members of $\mathcal{S}$:

$$\text{span}(\mathcal{S}) := \{X_1 \triangle X_2 \triangle \cdots \triangle X_k : X_1, \dots, X_k \in \mathcal{S}, k \in \mathbb{N}\}.$$

By convention, the symmetric difference of zero sets is $\emptyset$, so $\text{span}(\mathcal{S})$ always contains $\emptyset$.

$\mathcal{S}$ is *dependent under* $\triangle$ if, for some distinct $X_1, \dots, X_k \in \mathcal{S}$ (i.e. $X_i \neq X_j$ for $i \neq j$), $X_1 \triangle \cdots \triangle X_k = \emptyset$. If $\mathcal{S}$ is not dependent under $\triangle$, it is *independent under* $\triangle$.

$\mathcal{S}$ is a *circuit under* $\triangle$ if it is dependent under $\triangle$ and also minimal with respect to that property. $\mathcal{S}$ is a *base under* $\triangle$ if it is independent under $\triangle$ and also maximal with respect to that property.

- Let $\mathcal{S}$ be linear under $\triangle$. Prove, by induction on k , that for all k and all $X_1, \dots, X_k \in \mathcal{S}$, $X_1 \triangle \cdots \triangle X_k \in \mathcal{S}$.
- Prove that, for every set $\mathcal{S}$ of subsets of E , $\mathcal{S}$ is linear under $\triangle$ if and only if $\mathcal{S} = \text{span}(\mathcal{S})$.
- Prove by induction on k that for all k and all $X_1, \dots, X_k \in \mathcal{S}$, either $X_1 \triangle \cdots \triangle X_k = \emptyset$, or there exist distinct $Y_1, \dots, Y_n \in \mathcal{S}$ with $n \leq k$ such that $X_1 \triangle \cdots \triangle X_k = Y_1 \triangle \cdots \triangle Y_n$.
- Prove that $\mathcal{S}$ is dependent under $\triangle$ if and only if there exists $X \in \mathcal{S}$ such that $X \in \text{span}(\mathcal{S} \setminus \{X\})$.
- Prove that if $\mathcal{S}$ is a minimal dependent set under $\triangle$, then for every $X \in \mathcal{S}$ we have $X \in \text{span}(\mathcal{S} \setminus \{X\})$.
- Prove that every minimal spanning set under $\triangle$ is independent under $\triangle$.
- Prove that every minimal spanning set under $\triangle$ is a maximal independent set under $\triangle$.
- Prove that every maximal independent set under $\triangle$ is also a spanning set under $\triangle$.

- (i) Prove that every maximal independent set under $\triangle$ is also a minimal spanning set under $\triangle$.
- (j) Prove that $\mathcal{S}$ is a minimal spanning set under $\triangle$ if and only if it is a maximal independent set under $\triangle$.
- (k) Prove that, if $\mathcal{S}$ is independent under $\triangle$ then $|\text{span}(\mathcal{S})| = 2^{|\mathcal{S}|}$.
- (l) Prove the independence augmentation property: if $\mathcal{S}$ and $\mathcal{T}$ are independent under $\triangle$ and $|\mathcal{T}| = |\mathcal{S}| + 1$, then there exists $X \in \mathcal{T} \setminus \mathcal{S}$ such that $\mathcal{S} \cup \{X\}$ is independent under $\triangle$.
- (m) Prove that all bases under $\triangle$ have the same size.
- (n) Do all minimal dependent sets under $\triangle$ have the same size? Give a proof of your claim.
- (o) Prove that, for every base $\mathcal{B}$ of $\mathcal{S}$ under $\triangle$, and every $X \in \mathcal{S}$, there is a unique way to write X as a symmetric difference of members of $\mathcal{B}$, using each element of $\mathcal{B}$ at most once.

For each previous part (a)–(o), state what type of proof you have used.

Finally: design a simple algorithm for the following problem — given a collection $\mathcal{S}$ of subsets of E with a nonnegative real weight for each member, output a subset $\mathcal{X}$ of $\mathcal{S}$ that is independent under $\triangle$ such that the sum of the weights of all members of $\mathcal{X}$ is maximum. Does this algorithm always find the best possible solution? Justify your answer, using a previous exercise.

Solution. (a)

■

Solution. (b)

■

Solution. (c)

■

Solution. (d)

■

Solution. (e)

■

Solution. (f)

■

Solution. (g)

■

Solution. (h)

■

Solution. (i)

■

Solution. (j)

■

Solution. (k)

■

Solution. (l)

Solution. (m)

■

Solution. (n)

■

Solution. (o)

■

Solution. Algorithm & justification

■

■

Exercise 13

Identify the errors in the following “theorem” and “proof”, which are both incorrect. The steps in the “proof” are numbered for convenience.

“Theorem.” For all $n \in \mathbb{N}$, a circular disc can be cut into 2^n pieces by n straight lines.

“Proof.”

1. Consider $n = 0$. The disc is not cut at all, so it is still in one piece. So the number of pieces is $2^0 = 1$. So the claim is true for $n = 0$.
2. Consider $n = 1$. A single line across a disc cuts it into two pieces, so the number of pieces is 2^1 . So the claim is true for $n = 1$.
3. Now consider $n = 2$. If the circle is cut by one line into two pieces, then we can cut it again by a line that crosses the first line. Such a line must cut right across both of the pieces created by the first cut. Therefore each of those pieces is divided in two, so the total number of pieces is $2 \times 2 = 2^2$, so the claim is true for $n = 2$.
4. It is clear from the cases considered so far that, once the circle is cut by some number of lines, then another line can be used to divide every piece into two pieces, thereby doubling the number of pieces.
5. So, if we use n lines, then the repeated doubling gives 2^n pieces. Therefore the claim is true for all n . “□”

Solution.

■

Exercise 14

Identify the errors in the following “theorem” and “proof”, which are both incorrect.

“Theorem.” Every rational number is an integer.

“Proof.”

1. We need to show that the sets $\mathbb{Z}$ and $\mathbb{Q}$ are equal.
2. To prove that two sets are equal, we can prove the appropriate subset and superset relations between the two sets.
3. We first prove the subset relation between these two sets, $\mathbb{Z} \subseteq \mathbb{Q}$.

4. Let $x \in \mathbb{Z}$.
5. Since x is an integer, we have $x = \frac{x}{1}$.
6. So x is a quotient of integers.
7. Therefore x is rational, i.e., $x \in \mathbb{Q}$.
8. So we have proved that $\mathbb{Z} \subseteq \mathbb{Q}$.
9. Now we need to prove the superset relation, which is the reverse of subset.
10. So we will prove that $\mathbb{Q} \supseteq \mathbb{Z}$.
11. Let $x \notin \mathbb{Q}$.
12. Then x cannot be written in the form $\frac{p}{q}$ where p and q are integers.
13. So it certainly cannot be written in this form with $q = 1$.
14. Therefore x is not an integer, i.e., $x \notin \mathbb{Z}$.
15. So we have proved that $\mathbb{Q} \supseteq \mathbb{Z}$.
16. It follows from our subset and superset relations between the two sets that $\mathbb{Z} = \mathbb{Q}$. “□”

“God made the integers; all else is the work of man.” — Leopold Kronecker (1823–1891)

Solution.

■

Exercise 15

Let G be the cryptosystem obtained from Caesar slide by restricting the keyspace to the first half of the alphabet (see Exercise 2.15 for definitions relating to the Caesar slide cryptosystem). So the keyspace K_G is

$$\{a, b, c, d, e, f, g, h, i, j, k, l, m\}.$$

Prove that G is not idempotent.

Solution.

■

A Vector spaces, independence, and dimension

This appendix collects the background from linear algebra used throughout Exercise 5: the idea of a vector space, what it means for a set of vectors to be independent, and why every basis of a given space has exactly the same number of vectors. The matroid axioms (I1)–(I3) used in Exercise 5 are, in a precise sense, what remains of this theory once the vectors themselves are stripped away and only the combinatorics of independence is kept.

A.1 Vector spaces

A *vector space* V is a set of objects, called vectors, that can be added together and multiplied (“scaled”) by numbers, in such a way that the usual rules of algebra hold: addition is commutative and associative, there is a zero vector $\mathbf{0}$ with $v + \mathbf{0} = v$ for every v , every v has a negative $-v$ with $v + (-v) = \mathbf{0}$, and scaling distributes over addition in the expected way. The space $\mathbb{R}^n$ of columns of n real numbers is the prototypical example, but the same rules govern spaces of polynomials, spaces of matrices, and spaces of functions.

A *subspace* of V is a subset $W \subseteq V$ that is itself a vector space using the same addition and scaling. Concretely, W is a subspace precisely when it contains $\mathbf{0}$ and is closed under addition and scaling: if $u, w \in W$ and c is a scalar, then $u + w \in W$ and $cu \in W$. Lines and planes through the origin in $\mathbb{R}^3$ are subspaces; a line or plane that misses the origin is not, since it fails to contain $\mathbf{0}$.

A.2 Linear combinations and span

Given vectors $v_1, \dots, v_k \in V$ and scalars $c_1, \dots, c_k$, the vector

$$c_1v_1 + c_2v_2 + \dots + c_kv_k$$

is called a *linear combination* of $v_1, \dots, v_k$. The set of *all* linear combinations of $v_1, \dots, v_k$ is called their *span*, written $\text{span}(v_1, \dots, v_k)$. The span is always a subspace of V : it is closed under addition (adding two linear combinations gives another linear combination) and under scaling. Geometrically, the span of a single nonzero vector in $\mathbb{R}^3$ is a line through the origin; the span of two vectors that do not lie on a common line is a plane through the origin.

A.3 Linear independence

The vectors $v_1, \dots, v_k$ are *linearly independent* if the only way to combine them to produce the zero vector is trivially:

$$c_1v_1 + c_2v_2 + \dots + c_kv_k = \mathbf{0} \implies c_1 = c_2 = \dots = c_k = 0.$$

If some nontrivial combination (not all c_i equal to zero) does produce $\mathbf{0}$, the vectors are *linearly dependent*. A dependent set always contains a redundant vector: if $c_1v_1 + \dots + c_kv_k = \mathbf{0}$ with, say, $c_1 \neq 0$, then

$$v_1 = -\frac{c_2}{c_1}v_2 - \dots - \frac{c_k}{c_1}v_k,$$

so v_1 can be discarded without shrinking the span. Independence is therefore best understood as the absence of such redundancy: an independent set is one in which every vector contributes something that the others cannot already produce.

Example A.1. In $\mathbb{R}^2$, the vectors $(1, 0)$ and $(0, 1)$ are independent: if $c_1(1, 0) + c_2(0, 1) = (0, 0)$, then $c_1 = c_2 = 0$. The vectors $(1, 1)$, $(2, 2)$ are dependent, since $2(1, 1) - (2, 2) = (0, 0)$ is a nontrivial combination equal to zero; indeed $(2, 2)$ is simply twice $(1, 1)$ and contributes nothing new.

An elementary but important fact is that any set of vectors containing $\mathbf{0}$ is automatically dependent: taking $c_1 = 1$ for the zero vector and $c_i = 0$ for all others gives a nontrivial combination summing to $\mathbf{0}$.

A.4 Basis

A *basis* of a vector space V is a set of vectors that is simultaneously:

- (i) linearly independent, and
- (ii) spanning, meaning every vector in V is a linear combination of the basis vectors.

Independence guarantees there is no redundancy in the basis; spanning guarantees nothing is missing. Together, these two conditions mean that every vector in V can be written as a linear combination of the basis vectors in *exactly one* way — this is the coordinate representation of a vector relative to that basis.

Example A.2. The *standard basis* of $\mathbb{R}^n$ consists of the vectors $e_1, \dots, e_n$, where e_i has a 1 in position i and 0 elsewhere. Every vector $(a_1, \dots, a_n) \in \mathbb{R}^n$ is uniquely $a_1e_1 + \dots + a_ne_n$.

A given space almost always has infinitely many different bases — for example, $\{(1, 0), (1, 1)\}$ is also a basis of $\mathbb{R}^2$, distinct from the standard basis $\{(1, 0), (0, 1)\}$. What is remarkable, and what makes the notion of dimension possible, is that no matter which basis is chosen, it always has the same number of vectors.

A.5 Dimension

Theorem A.3 (Invariance of basis size). *Any two bases of a vector space V have the same number of vectors.*

The proof rests on a single combinatorial fact, the *exchange lemma*: if A is a linearly independent set and every vector in A can be written as a linear combination of the vectors in a set B , then A cannot have more vectors than B . Briefly, one exchanges vectors of B for vectors of A one at a time — each exchange is justified because some vector of B appears with a nonzero coefficient in the expression for the next vector of A , so that vector of B can be solved for and replaced. If A had more vectors than B , this process would run out of vectors in B to exchange before all of A had been used, forcing some later vector of A to be a combination of earlier vectors of A — contradicting the independence of A .

Applying the exchange lemma twice — once with A and B playing the roles of two different bases, and once with their roles reversed, since each basis spans V and so can express every vector of the other — shows that neither basis can have more vectors than the other, so they must have exactly the same number of vectors. This common number is called the *dimension* of V , written $\dim V$.

Example A.4. $\dim \mathbb{R}^n = n$, witnessed by the standard basis $\{e_1, \dots, e_n\}$. A plane through the origin in $\mathbb{R}^3$ has dimension 2; a line through the origin has dimension 1; the space $\{\mathbf{0}\}$ containing only the zero vector has dimension 0, since its only basis is the empty set.

A.6 From vector spaces to matroids

The theory above depends, at first sight, essentially on the algebraic structure of a vector space: addition, scalar multiplication, and the ability to solve $c_1v_1 + \dots = \mathbf{0}$ for one vector in terms of the others. Yet inspecting the proof of the invariance of basis size shows that only two facts about independence were actually used:

- (i) every subset of an independent set is independent, and

- (ii) the exchange property: given two independent sets of different sizes, some element of the larger one can be added to the smaller one while preserving independence.

No reference to vectors, scalars, or linear combinations survives in the final argument. This observation — that the essential content of “independence” and “dimension” can be isolated from the vector-space setting entirely — is what led Whitney (1935) to define a *matroid*: a finite set E together with a collection $\mathcal{I}$ of “independent” subsets satisfying exactly properties (I1), (I2), and (I3) of Exercise 5. Under this abstraction, a basis becomes a maximal independent set, dimension becomes the common size of all maximal independent sets, and the exchange lemma becomes axiom (I3) itself, assumed outright rather than derived. Exercise 5 develops this abstract theory directly from the axioms, and Part (c) in particular is the matroid-theoretic proof of the invariance of basis size given above, with every reference to vectors removed.